American Computer Science League

2020-2021 ● Contest 4: Solutions ● Elementary Division

1. Graph Theory The set of undirected edges for the entire graph is {AC, AD, AE, AF, BC, BD, BE, CD, CE, CF, DF}. The girl with the most friends is the letter used the most. **Carol** has a total of 5.	**1.** Carol (C)
2. Graph Theory A complete graph includes an edge from every vertex to every other vertex. There should be 4+3+2+1=10 edges. The current graph has 7 edges so **3** more are needed. They are: QR|RQ, RT|TR, PS|SP.	**2.** 3 (C)
3. Graph Theory By inspection, the **7** simple paths from vertex A to vertex D are: AD, ACD, AFD, ABCD, ABFD, AFBCD, and ACBFD.	**3.** 7 (D)
4. Graph Theory There are 2 of length 3: ADEA, AEDA. There are 2 of length 4: ABCEA, AECBA. There are 2 of length 5: ABCEDA, ADECBA. There are a total of **6** of them. The number of cycles should be even.	**4.** 6 (B)
5. Graph Theory Using Euler's rules, the number of vertices of odd degree must be either 0 or 2 for a graph to be transversable. The degree of each vertex is: C=2, D=4, E=2, F=2, G=3, H=2, J=3. There are 2 which have an odd number of edges: G and J so the answer is **GJ**.	**5.** GJ (A)